CHAPTER 3
METHODOLOGY

3.1 WORKFLOW OF THERMAL CONDITION RECOGNITION

The thermal images conditions recognition has begin with collecting dataset by thermal camera [5]. The thermal cameras are used to detect equipment temperature differences indicating heat patterns compared to regular cameras as they work based on visible light. The project workflow is shown in Fig 3.1. The images are collected and they are organised [1]. They are labeled based on their conditions as normal or abnormal. This labeling is more effective as it trains the machine learning model with the labeled data. Also this dataset is preprocessed . As they need to be resized and enhanced so that the image quality is improved and they can be more useful during training the model. It ensures that there is uniformity across the dataset and allots the model to analyze more efficiently. After this step their features are extracted.

In feature extraction defined patterns are identified from the given thermal image. extraction is done based on two types of features .One is deep feature which is extracted using EfficientNet-B0 model [17], complex image patterns are captured in this deep learning architecture. In addition, this texture feature is also obtained using Local Binary Patterns (LBP) method [13], this focuses on recognizing fine texture details and temperature variations. These two features that is deep and texture dataset enable the design of a hybrid model. These datasets are used to train the hybrid model to learn different thermal conditions. The models are tested and validated to check their performance metrics.

Finally, the trained system is used for thermal condition recognition by analyzing new thermal images and automatically identifying their operating condition. This allows for efficient detection of abnormalities or faults. To further enhance interpretability, the predictions are analyzed using an Explainable AI (XAI) framework. Grad-CAM highlights the important regions in the thermal image that influence the model decision [7], [8], while SHAP explains the contribution of individual features [19], [20]. This explanation was then presented in a human-readable form, making the system more transparent and practically useful.

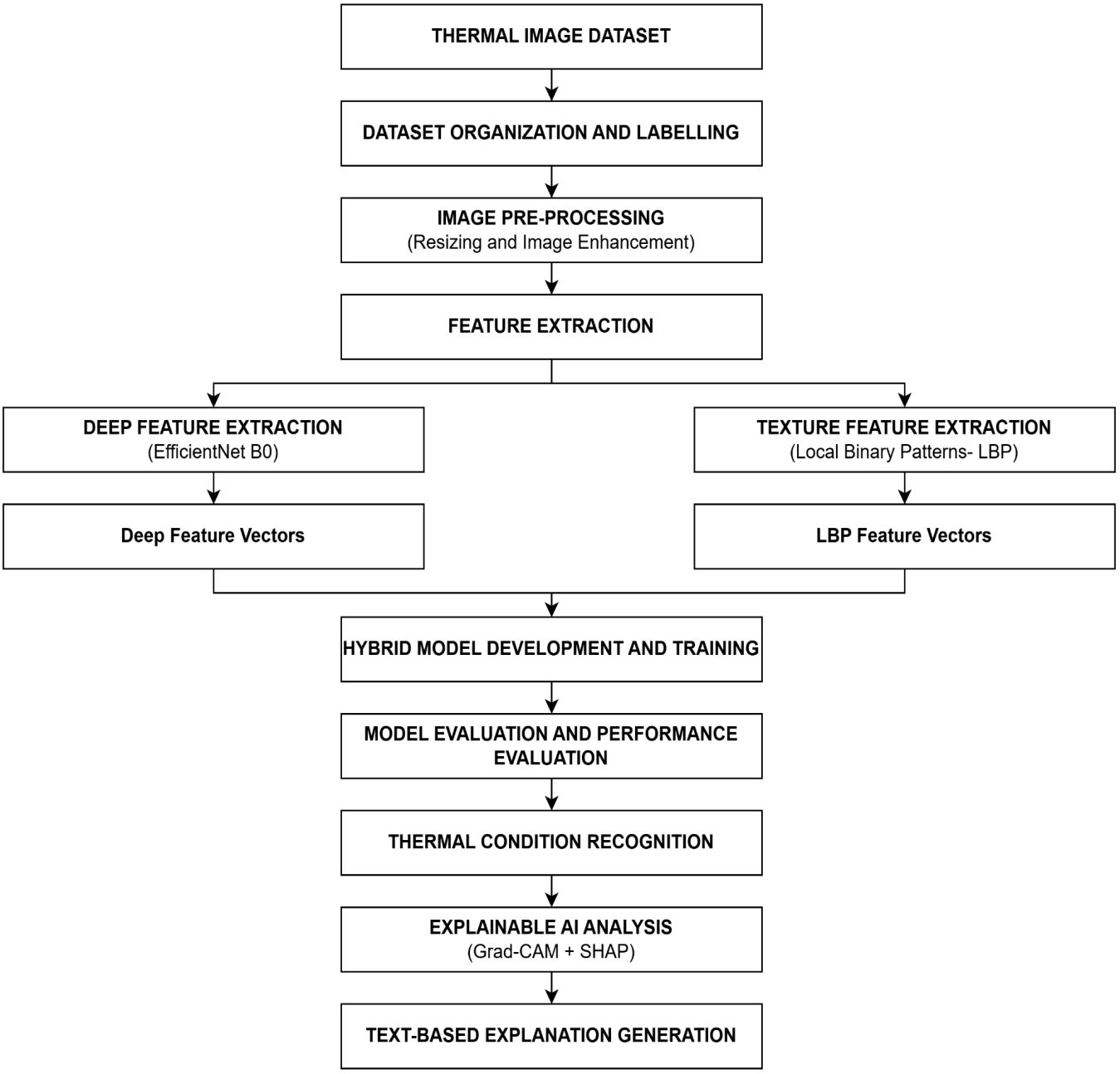

Fig 3.1: Flow Diagram of Thermal Condition Recognition

3.2 DATASET DESCRIPTION

The present study makes use of a publicly available infrared thermal image dataset that contains images of high-voltage electrical equipment. This dataset is used to develop and evaluate the proposed recognition model [8]. Identifying components in high-voltage substations accurately is very important for monitoring the health of power systems. It helps in detecting possible faults at an early stage while the equipment is still functioning, which in turn supports reliable measurement, control, and protection of the electrical network.

The dataset includes thermal images of different types of high-voltage equipment typically found in 132 kV air-insulated substations. These images were captured using a portable FLIR C5 infrared camera at different times and under various load conditions to reflect real operating environments. Unlike normal images that depend on visible light, thermal images represent temperature distribution across the surface of the equipment. Because of this, they are not affected much by lighting conditions and are highly useful for identifying abnormal heat patterns that may indicate faults [2].

There are five categories of equipment in the dataset: circuit breakers, power transformers, surge arresters, disconnectors, and wave traps. The dataset contains 203 images of circuit breakers, 178 of power transformers, 181 of surge arresters, 180 of disconnectors, and 153 of wave traps. Each image has a resolution of 640 × 480 pixels and is stored in RGB format using a rainbow color scheme to represent different temperature levels. A color bar is also included in each image to indicate the temperature range during capture. The images are well-organized into labeled folders, making them suitable for training and testing machine learning models.

One key advantage of this dataset is that it contains multiple types of equipment, allowing models to learn and differentiate between their unique thermal characteristics. Since infrared imaging captures long-wave radiation, it reflects surface temperature variations that are closely related to the condition and performance of the equipment. Additionally, the dataset includes images collected under different environmental and operational conditions, which helps improve the model's ability to perform well in real-world scenarios.

Overall, the dataset is publicly available and provides enough variety and data samples for developing computer vision and deep learning models. In this work, it is used to extract deep features using convolutional neural networks [17], [18], along with handcrafted texture features. These features are then combined using a hybrid approach to enhance the accuracy of equipment classification and support effective fault detection and monitoring.

3.3 HIGH VOLTAGE SUBSTATION EQUIPMENT

Power Transformers

Power transformers are essential components in electrical substations, as they are used to increase or decrease voltage levels for efficient power transmission and distribution. These devices operate continuously and are subjected to significant electrical and thermal stress during operation. Thermal imaging of transformers reveals how heat is distributed across different parts such as the windings, outer tank, and cooling radiators. This temperature information is highly useful for monitoring the condition of the transformer and detecting possible faults at an early stage.

Surge Arrestors

Surge arresters play an important role in protecting substation equipment from sudden voltage surges caused by lightning strikes or switching actions. They work by safely diverting excess voltage to the ground. Thermal imaging can be used to observe temperature variations in these devices, helping to identify unusual heating patterns. Such abnormalities may be due to leakage currents or internal wear, which can signal a potential failure if not addressed in time.

Circuit Breakers

Circuit breakers are important switching devices used to stop fault currents and isolate damaged parts of a power system. During operation, they experience high electrical stress, which can sometimes lead to the formation of thermal hotspots due to issues like contact resistance or internal faults. Infrared imaging helps in detecting these temperature variations, making it easier to diagnose potential problems at an early stage [3].

Wavetraps

Wave traps, also known as line traps, are placed on transmission lines to block high-frequency communication signals used in power line carrier (PLC) systems. Their main function is to keep these signals confined to specific sections of the network. Thermal imaging can help identify unusual heat patterns in wave traps, which may be caused by abnormal current flow or insulation problems.

Disconnectors

Disconnectors (isolators) are mechanical switching devices used to isolate equipment for maintenance purposes. Unlike circuit breakers, they are operated under no-load conditions. Thermal imaging is useful for identifying contact resistance problems, loose connections, or abnormal heating at joints.

3.4 THERMAL IMAGE ACQUISITION USING FLIR C5 CAMERA

The thermal images used in this dataset were obtained with the FLIR C5 thermal imaging camera, a compact handheld device commonly used for industrial inspection, electrical maintenance, and condition monitoring tasks [8]. This camera was employed to capture the temperature distribution on the surfaces of various substation components, including transformers, surge arresters, circuit breakers, wave traps, and disconnectors.

One of the key features of the FLIR C5 is its radiometric capability, which allows each pixel in the image to store accurate temperature data. This is important for identifying hotspots and unusual heating patterns, detecting equipment operating under thermal stress, and supporting effective feature extraction for machine learning models. It also helps maintain consistent data quality even when environmental conditions change.

In addition, the camera's compact and portable design made it convenient for on-site data collection in active substation environments. This ensured that the captured images closely represent real operating conditions, improving the reliability of the dataset.

3.5 LOCAL BINARY PATTERNS

Local Binary Pattern (LBP) is a simple and powerful feature extraction technique used in image processing and computer vision for describing the texture of an image [11].The primary idea behind LBP is to encode the local spatial structure of an image by comparing each pixel with its neighboring pixels and representing the result as a binary number [13]. This method transforms an image into a set of numerical values that effectively capture local texture patterns.

In LBP, each pixel in a grayscale image is processed by considering a small neighborhood around it, typically a 3×3 window. The center pixel is taken as a reference, and its intensity value is compared with those of its surrounding eight neighbors. For each neighboring pixel, a binary value is assigned based on the comparison:

If the neighbor's intensity is greater than or equal to the center pixel, it is assigned a value of 1; otherwise, it is as 0.

The LBP operator can be mathematically expressed as:

$$LBP(x_c, y_c) = \sum_{p=0}^{P-1} s(g_p - g_c).2^P \quad (1)$$

Where:

(x_c, y_c)= coordinates of center pixel

g_c= gray level of center pixel

g_p= gray level of neighboring pixels

p = number of neighbors

s(x) is defined as:

$$s(x) = \{1, x \geq 0 \quad (2)$$

and

$$s(x) = \{0, x < 0 \quad (3)$$

This equation converts local intensity differences into a binary code.

These binary values are arranged in a sequence, usually in a clockwise direction, forming an 8-bit binary number. This binary number is then converted into its decimal equivalent, which becomes the LBP value for that pixel. By repeating this process for every pixel in the image, a new image known as the LBP image is generated.

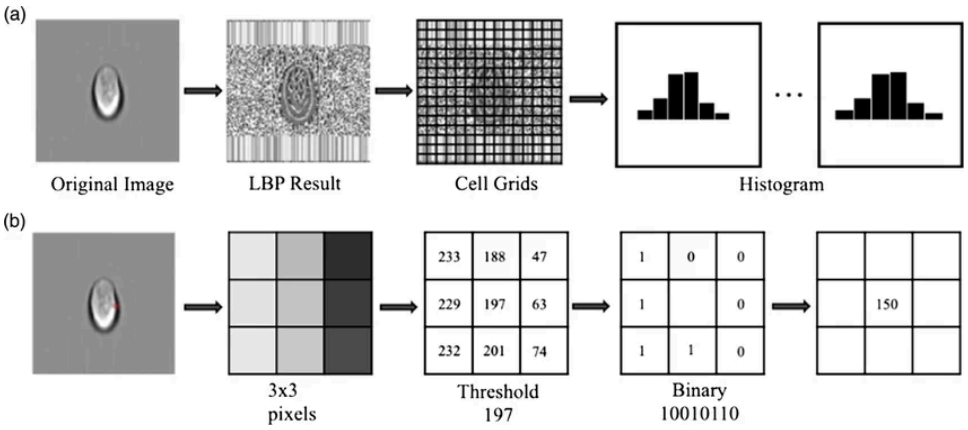

Fig 3.2 LBP Analysis

The working process of the Local Binary Pattern (LBP) technique for extracting texture features is shown in Fig. 3.3. In part (a), the method starts with a grayscale image, which is then converted into an LBP image by comparing each pixel with its neighboring pixels. This comparison helps in capturing local texture patterns. The generated LBP image is further divided into smaller regions or grids, and a histogram is calculated for each grid. These histograms represent the distribution of binary patterns and together form a strong feature descriptor for the image.

Part (b) explains the process in more detail. A 3×3 window is selected, where the center pixel (for example, 197) is taken as a reference. Each surrounding pixel is compared with this value. If the neighboring pixel value is greater than or equal to the center, it is assigned a value of 1; otherwise, it is assigned 0. This results in a binary pattern (for example, 10010110), which is then converted into a decimal number (such as 150). This number is added to the histogram, helping to represent the texture information of the image.

One of the main advantages of LBP is that it is simple and computationally efficient, making it suitable for real-time applications and systems with limited resources. It is also

robust to changes in lighting conditions because it is invariant to monotonic grayscale variations [14], [15], [16]. Moreover, LBP does not require complex preprocessing steps or parameter tuning, which makes it easy to apply in practical scenarios.

However, LBP has some limitations. It is sensitive to noise, especially in areas where pixel intensity values are very close to each other. Even small changes in pixel values can alter the binary pattern and affect the results [24]. In addition, LBP mainly captures local texture features and does not account for the overall structure of the image, which can reduce its effectiveness in more complex situations. To address this issue, EfficientNet-B0 is incorporated into the model to capture higher-level and global features, thereby improving overall performance.

3.6 EFFICIENT NET B0

EfficientNet-B0 is a convolutional neural network architecture designed for image classification that belongs to the EfficientNet family [17]. It is known for achieving high accuracy while maintaining low computational complexity, making it highly efficient compared to traditional deep learning models. EfficientNet-B0 serves as the baseline model in the EfficientNet series and is built using a systematic scaling method that balances network depth, width, and resolution [10].

The core idea behind EfficientNet-B0 is the concept of compound scaling, which uniformly scales all dimensions of a neural network rather than arbitrarily increasing one aspect such as depth or width [19]. Traditional models often increase layers or channels independently, which can lead to inefficiencies and overfitting. EfficientNet-B0 addresses this issue by using a compound coefficient that scales depth, width, and input image resolution in a balanced manner, resulting in better performance with fewer parameters.

Compound Scaling Formula:

EfficientNet introduces a compound scaling method that uniformly scales the network's depth, width, and input resolution using a fixed set of scaling coefficients.

The scaling is defined as:

$$d = \alpha^{\phi} \tag{4}$$

$$w = \beta^{\phi} \tag{5}$$

$$r = \gamma^{\phi} \tag{6}$$

Where:

- d = network depth (number of layers)
- w = width (number of channels)
- r = input image resolution
- α,β,γ = constants determined experimentally
- ϕ = scaling coefficient (controls model size)

Constraint:

$$\alpha \cdot \beta^2 \cdot \gamma^2 \simeq 2 \tag{7}$$

When the model size is increased, all dimensions are scaled proportionally, not randomly.

One of the key advantages of EfficientNet-B0 is its efficiency in terms of accuracy versus computational cost [17]. It achieves competitive performance on benchmark datasets like ImageNet while using significantly fewer parameters compared to traditional architectures such as ResNet and VGG [18]. This makes it particularly suitable for deployment in resource-constrained environments such as mobile devices, embedded systems, and real-time applications.

3.7 EXPLAINABLE AI (XAI)

Explainable Artificial Intelligence (XAI) focuses on making machine learning models more transparent by providing clear reasoning behind their predictions. This is especially important for complex models like convolutional neural networks (CNNs), where the internal decision-making process is not easily understood. Although such models perform very well in image-based tasks, they often operate as black boxes. XAI methods help bridge this gap by linking the model's outputs with the input features, allowing us to verify whether the predictions are based on meaningful and relevant patterns [13].

3.7.1 Types of Explainability

Explainability techniques can generally be divided into two categories: intrinsic and post-hoc methods. Intrinsic models, such as decision trees, are naturally interpretable because their structure is easy to follow. However, they are not well-suited for handling complex image data. For this reason, post-hoc methods are commonly used with deep learning models.

Post-hoc approaches can be further grouped into two types:

- Model-agnostic methods:
  These approaches treat the model as a black box and generate explanations without relying on its internal structure. One widely used technique is SHAP, which is based on game theory. It assigns importance values to input features by estimating how much each feature contributes to the final prediction.
- Model-specific methods:
  These method takes advantage of the internal working of the model. Grad-CAM is a popular example designed specifically for CNNs. It produces visual maps that highlight the important regions in an image that influence the model's decision.

3.7.2 XAI Architectural Integration

In the proposed system, XAI is incorporated as a structured three-stage pipeline consisting of a prediction model, an explanation module, and a visualization layer. This framework is integrated into the overall fault detection process [20], [21].

The prediction model is based on EfficientNet and is trained using FLIR thermal images. It performs both feature extraction and classification, producing probabilities that indicate whether the equipment condition is normal or abnormal.

The explanation module works alongside the trained model without altering its structure. It uses both Grad-CAM and SHAP to interpret the predictions. Grad-CAM identifies important spatial regions by analyzing gradients from the final convolutional layer, while SHAP calculates how much each feature contributes to the output [25].

The visualization layer presents these results in an understandable form by overlaying heatmaps on the original thermal images. This helps in directly identifying critical areas, such as hotspots, that influence the model's decision.

3.7.3 Implementation in Thermal Fault Detection

In this study, XAI is applied to an EfficientNet-based model trained on FLIR thermal images to classify equipment conditions as either normal or abnormal. Since the model mainly provides probability outputs, additional interpretation is required to understand its predictions.

Two complementary techniques are used:

- Grad-CAM (Spatial Explanation):
  This method generates heatmaps by analyzing gradients related to the target class. These heatmaps highlight the regions in the thermal image that have the greatest influence on the prediction.
- SHAP (Feature Contribution):
  SHAP assigns importance values to different parts of the input, indicating how each region increases or decreases the likelihood of a particular classification.

3.7.4 Role of XAI in the Proposed System

Within the overall system, XAI acts as a validation and analysis tool. It helps to:

- Confirm that the model focuses on meaningful thermal regions
- Identify fault locations through hotspot detection
- Measure the contribution of different features
- Analyze incorrect predictions to improve model performance

By incorporating XAI, the system becomes not only accurate but also more trustworthy and easier to interpret, which is essential for real-world applications [12], [21].

Finally, the thermal condition recognition framework combines thermal imaging with advanced machine learning techniques to effectively detect and classify temperature-related patterns in equipment. The process involves careful data collection, proper labeling, preprocessing for uniformity, and feature extraction using both EfficientNet-B0 and LBP methods. By combining deep and texture features, a strong hybrid model is developed that can capture both complex patterns and fine details. Through proper training, validation, and evaluation, the system achieves reliable performance and can analyze new thermal images to automatically detect faults or abnormalities. Overall, this approach improves efficiency, supports predictive maintenance, and enhances safety and operational reliability.